

The Fundamentals of Programming

Strings



Objectives

What you will learn:

1. How to use different string functions.
2. How Python represent strings and parts of strings.
3. To use splicing and string operators to complete complex tasks.

Strings

The basics

Strings in Python work a lot like integers and floats in the way you can deal with them.

You can add

```
s1 = 'Hi'  
s2 = 'You!'  
s3 = s1 + ' ' + s2
```

You can multiply

```
s1 = 'Hi'  
s2 = s1 * 5
```

You can't subtract or divide

```
s1 = 'Hi'  
s2 = 'You!'  
s3 = s1 / s2  
s3 = s1 - s2
```

Using Strings

Strings have many special functions that can be used with, though many are used in a different way than the functions you've seen so far.

```
var = 'the blue dog'
```

Length:

```
print(len(var))
```

Count:

```
print(var.count('e'))
```

Uppercase:

```
print(var.upper())
```

Is it a digit:

```
print(var.isdigit())
```

Practice

Using strings

Determine the output from the following programs. You may search for the functions used.

```
var = 'Hello'  
var += '-Bye'  
print(var.lower())
```

```
var = 'Hello'  
var = var.upper()  
var = var.capitalize()  
print(var, str(len(var)))
```

```
var = 'Hello'  
print(var.islower())  
var = var.count('O')  
print(var.isdigit())
```

What is a string?

Digging into the details

Strings work differently than other types of variables. As you've seen they have many different functions that can be used with them, but that's not all. Strings are actually broken up into individual **characters** each of which assigned an **index**.

var = "Hello"

0	1	2	3	4	Index
H	e	l	l	o	Element

var[1] is 'e'
var[4] is 'o'
var[0] is 'H'

String examples

Index and element

It's important to get the terms index and element correct, as they'll be used with lists as well. Remember that an index always starts at zero and you should be fine.

Answer the questions to the right.

0	1	2	3	4	5	6	7	8
A		w	o	r	d		t	o

1. What element is at index 7?
2. What is the index of the element t?
3. What element is at index 8?
4. What is the index of the element w?

Practice

Elements and indices

Using what know about indices and elements, try and figure out the output for the following programs.

```
var = 'Hello'  
print(var[2])
```

```
var = 'Hello'  
var += 'Bye'  
print(var[7])
```

```
var = 'Hello'  
print(var[10])
```


Cutting up Strings

You can access string elements by using the index and the square brackets `[ ]`. You can also use multiple values to take a splice of a string, which is section of one or more letters.

```
var = 'the blue dog'
```

Element:

```
print(var[2])
```

Splice from the middle:

```
print(var[1:4])
```

Splice from the start:

```
print(var[:5])
```

Splice until the end:

```
print(var[1:])
```

Practice

Splicing

Determine the output from the following programs. Remember to count only up to the last index, but not include it.

```
var1 = "abcd"  
var2 = "wxyz"  
print(var1[:2] + var2[1:3])
```

```
var1 = "abcd"  
var2 = "wxyz"  
print(var1[:] + var2[2:] + var1[:3])
```

Recap

We learned

1. How to access string elements using their index.
2. The different function available to manipulate strings.
3. How to splice strings.